QLOT: Queried Learned Optimization for Face Landmark Tracking

Dominik Gschwind Martin Weisenhorn Hannes Badertscher
Eastern Switzerland University of Applied Sciences (OST)
Rapperswil-Jona, Switzerland
{dominik.gschwind, martin.weisenhorn, hannes.badertscher}@ost.ch

Abstract

Facial landmark detection is a mature field, with tracking approaches capable of real-time inference on mobile devices. Yet most applications treat tracking as a post-processing problem, applying classical filters for jitter suppression. We propose QLOT (Queried Learned Optimization for Tracking), which treats face alignment as an iterative, query-conditioned learned optimization process rather than single-shot inference. Incorporating ideas from RAFT, a recurrent optical flow estimator, QLOT fuses per-landmark correlation features with temporal context carried by a recurrent hidden state, exchanged across landmarks through a low-rank Write-Mix-Read block. A composite objective of decoupled spatial and temporal Gaussian negative log-likelihood terms supervises directional uncertainty and the temporal derivatives of the error. QLOT achieves competitive accuracy to the state-of-the-art while improving temporal stability, staying efficient enough for real-time mobile deployment (2M parameters, 0.65G MACs/frame). To evaluate tracking, we generalize Normalized Mean Flicker to a multi-timescale stability profile based on the Allan variance.

1. Introduction

Facial landmark detection (FLD), the task of extracting precise coordinate positions for semantic facial features, is a mature field of research. Such systems already power many applications, including facial expression recognition, gaze estimation, and augmented reality. Approaches like FaceMesh [1], which extract not only 2D locations but even a dense 3D mesh of vertices, have been widely deployed. Thus, FLD models have become commonplace in mobile devices, while typically also requiring real-time performance.

Despite this commercial success, surprisingly few approaches [2, 3, 4, 5, 6] incorporate temporal information as a part of their end-to-end architecture. Instead, most methods strictly function as *pure detectors* that process continuous video streams as a series of independent frames. This stateless nature makes temporal instability or *jitter* virtually unavoidable. Visually, jitter manifests as high-frequency noise, where the predicted coordinates move erratically across subsequent frames, even though the underlying facial features remain static.

Large-capacity models, for example [7], can achieve relatively stable predictions through brute-force feature extraction, but their massive computational requirements make them infeasible for real-time mobile deployment. Practical detection systems designed for video-use thus rely on post-hoc classical filters [8, 9, 10] to dampen jitter. However, making use of temporal correlations as part of the architecture tackles the jitter problem directly. Moreover, using past alignments as structural priors can even increase detection performance.

We note five fundamental sources of jitter in FLD:

1. **Input instability.** Noise and instability of the cropped face region—typically from a separate model or tracking algorithm—propagate through the system and affect the landmark predictions [7].
2. **Label noise and inconsistencies.** Suboptimal annotation quality, semantic differences between datasets, or more generally, label noise, teach the model uncertain outputs that are input-sensitive [7, 11, 12].
3. **Geometric ambiguities.** Extracting the underlying 3D facial geometry from monocular 2D images is an ill-posed problem due to the scale ambiguity. Thus, perspective changes or occlusions can cause large position variations.
4. **Visual variability, noise and occlusions.** Changes in lighting, pose, expression, and self or external occlusions can introduce oscillations between plausible landmark configurations [5, 13, 14].
5. **Lack of temporal context.** A pure detector processes each frame independently, and so cannot utilize strong temporal correlations for antialiasing or noise suppression.

While post-hoc filtering stabilizes positions, it operates solely on output coordinates, causing oversmoothing and lag. Integrating a temporal pathway, learned end-to-end by training on video, can directly prevent this. Still, training on static images remains essential as annotated video data is scarce. A practical solution must satisfy the mobile performance budget, while bounding runaway error accumulation or *drift*—a critical concern for recurrent tracking.

Based on RAFT [15], a recurrent optical flow estimator, and [16], a query point-based FLD model, we design a stable recurrent architecture exhibiting near-state-of-the-art accuracy with mobile-friendly efficiency. Furthermore, we generalize a temporal stability metric based on the Allan variance, which allows us to quantify both jitter and drift. Our main contributions are as follows:

- We propose *QLOT*, Queried Learned Optimization for (Face Landmark) Tracking, a query-conditioned recurrent architecture that reformulates landmark detection as iterative learned optimization over time. It fuses semantic history and local image evidence through a low-rank *Write-Mix-Read* (WMR) block, while being temporally stable and lightweight enough for real-time edge deployment.
- We extend the simple Gaussian Negative Log-Likelihood (GNLL) loss used in [16, 17] to a mixture of decoupled spatial and temporal GNLL losses, using full 2D covariance representations to allow for directional uncertainty. A simple L2 loss is used for accuracy supervision.
- We demonstrate how to effectively train QLOT on a mixture of image and video data. Here, adopting the query mechanism of [16] proves highly synergistic, as it eliminates the need for label translation or normalization across diverse datasets.
- To properly evaluate continuous tracking performance, we introduce a generalization of the Normalized Mean Flicker (NMF) [5] metric based on the Allan variance. This provides a principled way to characterize the jitter and drift behavior of any tracking model.

2. Related Work

FLD architectures are broadly categorized by their output representation into three paradigms [14, 18]. *Direct coordinate regression* methods predict points directly; they are fast but often yield local structural imprecisions unless refined via cascaded networks [19]. *Heatmap-based* architectures preserve spatial context by predicting dense probability distributions. They are accurate but sensitive to outliers and computationally heavy [20, 21, 22]. Pushing them to sub-pixel precision required evolving beyond standard losses (MSE, L_1), which suffer from dilated responses and oscillatory convergence, to formulations like the *Adaptive Wing Loss* [22]. Finally, *face model-based* methods regress the parameters of statistical shape models. 3D Morphable Models (3DMMs) are preferred since they model the inherent 3D facial geometry, gracefully handling occlusions, and are less affected by texture and lighting variations [23]. Instead, they trade off fine-grained local accuracy to maintain rigid global geometry.

Learned Optimization for Sparse Prediction. Rather than inferring coordinates in a single forward pass, QLOT treats face alignment as a *learned optimization* process. Introduced to FLD by the Mnemonic Descent Method (MDM) [24], this approach has recently dominated dense matching tasks like optical flow via RAFT [15], which refines estimates by indexing a precomputed correlation volume using a Gated Recurrent Unit (GRU) [25]. While RAFT operates on dense pixel grids, QLOT extracts vision evidence from per-landmark localized correlation volumes. These volumes are also dense maps, but—being interpreted by a learned update operator—they can remain at low backbone resolutions, avoiding high-resolution decoding that makes heatmaps expensive. Unlike cascaded estimation, where each stage is an independent sub-network [19], the learned optimizer shares parameters across iterations and landmarks, emitting position *deltas* that are accumulated as in [15, 24].

Query-based and Continuous Architectures. Datasets annotate landmarks under incompatible schemes where topology and semantic meaning differ. Traditionally, this made multi-source training dependent on label translation, adding label noise while discarding some semantics [16]. Recent continuous and query-based architectures break this limitation. By prompting a network with 3D queries from a canonical face mesh [7, 16] or a structural prompt [26], these models learn to dynamically output the corresponding landmarks. Similarly, transformer-based aligners refine via *learned* landmark queries, conditioning patch-based coordinate updates [27, 28], a paradigm extended to tracking by 1DFormer [2] using temporal attention. These queries encode a learned landmark identity, which for QLOT is derived from an *encoding* of the canonical 3D points.

Probabilistic and Uncertainty-aware FLD. To tackle label noise, [17] uses fully synthetic data. They then estimate a 2D circular Gaussian per landmark, allowing the model to differentiate between hard or occluded, and easy in-view locations. QLOT builds on this idea by extending it to a full *directional* 2D covariance representation, and makes use of it within spatial [7, 16, 17] and temporal GNLL losses.

Jitter-free Tracking. Jitter, also called flicker or temporal instability, can be quantified by the Normalized Mean Flicker (NMF) [5], and evaluated on dedicated video benchmarks such as 300-VW [29] and WFLW-V [5]. Standard practices treat it as a post-processing problem, applying heuristic filters [8], Kalman filtering [10], learned smoothing [29], or even optical flow-based tracking [9]. However, these methods induce lag and discard the rich semantic history available naturally in video. Efforts to encode temporal priors natively include within-batch synthetic temporal augmentation [30], multi-view tuning [31], or directly incorporating recurrent/transformer modules [6, 32]. *Recurrence without Recurrence* (RwR) [5] uses fixed-point optimization for position estimation. Although trained only on static images, RwR can apply temporal smoothing at test time, while reusing previous frame predictions as initialization. QLOT also forwards

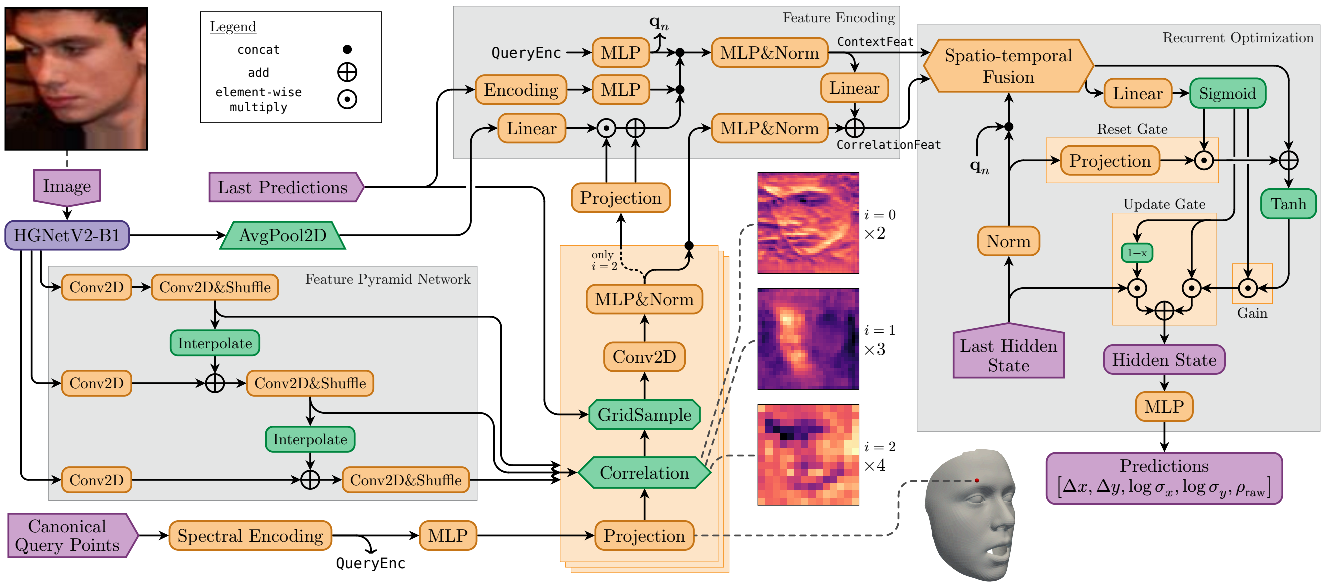

Figure 1. The QLOT architecture. Green components do not contain learned parameters. Most activation functions are omitted for clarity.

previous predictions, but learns actual temporal dynamics by propagating a recurrent hidden state and explicitly training via backpropagation through time.

Low-rank cross-element communication. Cross-element communication can avoid explicit *all*-pairs interactions by routing information through $S \ll N$ bottleneck elements,

$$\mathbf{Z} = \mathbf{B}_R \mathcal{M}(\mathbf{B}_W \mathbf{V}), \quad \mathbf{B}_W, \mathbf{B}_R^\intercal \in \mathbb{R}^{S \times N}, \tag{1}$$

where $\mathbf{B}_W$ writes values into the bottleneck, $\mathcal{M}$ mixes them, and $\mathbf{B}_R$ reads the result back. Linformer [33] uses a static learned write projection, identity mixing, and a query-to-projected-key attention matrix for the read. Set Transformer's ISAB encoder [34] first attends from learned inducing points to the elements and then from the elements to the induced features. Slot Attention [35] implements only the first direction. Competing slots aggregate inputs and are iteratively updated without elementwise readback. Perceiver IO [36] writes inputs to a latent array by cross-attention, mixes it with a deep latent transformer, and reads arbitrary output queries by cross-attention. Our WMR block predicts write and read routes *asymmetrically* from distinct signals, bottleneck slots are weighted *moment statistics* rather than attention outputs, and temporal information enters through routing, keeping the slots stateless unlike [35].

3. Method

In this section, we introduce our proposed QLOT framework, shown in Figure 1, including architectural details and training procedure. Then, we discuss a specific temporal stability metric and propose a physically motivated generalization which we use in Section 4. to evaluate jitter and drift.

3.1. Queried Learned Optimization for Tracking

QLOT reformulates facial landmark detection as an iterative, query-conditioned refinement process. It is composed of five core components: a lightweight *backbone*, *canonical query point* encoding, landmark-specific *correlation volume* generation, a *feature encoding* stage, and a *recurrent optimization* module.

3.1.1. Feature Extraction

As a state-of-the-art backbone in both quality and efficiency, we adopt *HGNetV2* [37]. An ImageNet-pretrained B1-variant receives square RGB images of size 224×224 pixels, outputting four levels of feature maps with strides 4, 8, 16, and 32 relative to the input image. We trim the last two levels to 384 and 192 channels, correcting channel-inflation for the original classification task, and removing 60% of the backbone parameters with a 10% reduction in FLOPs. Towards semantically rich features, we construct a *Feature Pyramid Network* (FPN) [38] of the first three levels using lightweight grouped and depthwise convolutions, simultaneously projecting to a common dimension of 96 channels. The final stride-32 map is average pooled and projected to a 96-dimensional *global image context* vector.

3.1.2. Canonical Query Points

3D canonical points [16], as a proxy of landmark identity, are annotated once on a shared reference mesh for all landmarks. The predictor is then constructed such that conditioning on any desired combination of these 3D points causes it to output only the corresponding landmarks. In our case, [39] is chosen as the canonical face shape (expression parameters $e_{24} = 0.5$ and $e_{26} = 0.4$, others zero). Like this, a single model can be trained jointly on multiple datasets with even heterogeneous annotations, while queries close the semantic gap between differing label schemes. At test time, any desired subset of keypoints can be extracted without retraining. Anchored to real 3D facial geometry, the canonical points also act as a structural prior over the spatial layout of facial features.

To obtain landmark identity features, each canonical query point is—after spectral encoding—fed through two 2-layer MLPs. The first, higher capacity MLP outputs a 192-dimensional vector that is fed into the *correlation volume* block. A second MLP produces an encoding of size 96 used as query-conditioning for the *recurrent optimization* module.

Spectral Encoding. Each canonical query point $\mathbf{p} \in \mathbb{R}^3$ is encoded by a phase-modulated positional encoding (PMPE),

$$\gamma(\mathbf{p}) = [\mathbf{p}, \sin(\omega_1 \mathbf{p} + \varphi_1), ..., \sin(\omega_L \mathbf{p} + \varphi_L)], \quad (2)$$

where frequencies ω_l and phases φ_l are *learnable*. The initialization provides a well-spread, aperiodic starting point. It combines a *Fourier branch* of 16 sin/cos pairs with geometric frequencies $\omega_k = \pi\tau^{-k/16}$, $\tau = 0.03$ [26], and a *phase-modulation branch* of 8 pairs covering the complementary low-frequency band [40]. In contrast to the original PMPE, we assign each phase-modulation channel a distinct carrier in $[\pi/4, \pi]$ from a golden-ratio Weyl sequence, and concatenate the two branches instead of summing, to leave the downstream MLP free to mix a total of 147 channels.

3.1.3. Correlation Volume

RAFT [15] uses pairs of frames to compute a dense 4D correlation volume, the same concept is applied to landmark-specific template matching. We produce a multi-level multi-head correlation volume for each landmark. The templates, or correlation kernels, stem from a learned per-level projection of the 192-dimensional query point-encoding. Each map level i—with feature map resolution 56×56 for $i = 0$, 28×28 for $i = 1$, and 14×14 for $i = 2$—is convolved with template kernels of size $R_i{\times}R_i$ in groups of K_i *correlation heads* giving K_i single-channel correlation maps per level.

Each feature map level, broadcast across N landmarks, has the shape $(N, K_i, 96/K_i, H_i, W_i)$ with square $R_i{\times}R_i$ kernels. This gives a computational complexity of $O(NR_i^2CH_iW_i)$ per level i where $C = 96$ is the total channel dimension. We experiment with different kernel sizes and group counts, but find that $R_i = 2, 1, 1$ and $K_i = 2, 3, 4$ give good results while keeping the cost of the expensive correlation operation low. Note that $R = 1$ simply computes a channel-wise dot product. In total, the correlation volume consists of shape (K_i, H_i', W_i') maps for each level i and landmark n. Correlation map resolutions $H_i'{\times}W_i'$ are the result of valid-padded convolutions. The volume is computed once per input image since neither the query points nor the backbone features change across refinement iterations.

3.1.4. Feature Encoding

Every iteration, the correlation volume is indexed around the previous landmark estimates. A 5×5 grid centered on these coordinates bilinearly samples one patch per head k, level i, and landmark n. Noisy but weak correlation responses are dampened by a learned-temperature soft-shrink activation inspired by [41],

$$\sigma(x) = x \cdot \mathrm{sigmoid}\big(|x|/\tau_{k,i}\big). \quad (3)$$

Instead of flattening, grouped 3×3 level-specific convolutions—exploiting the persisting spatial structure—extract intermediate features (126, 135, and 144 channels for levels $i = 0, 1, 2$, respectively). After projection and Root-Mean-Square (RMS) normalization, all levels are concatenated and compressed with a 2-layer MLP followed by another RMS norm to form the initial *correlation feature vector* $\mathbf{f}_n'$.

For stable visual, geometric and temporal grounding we construct *context features* $\mathbf{c}_n$ from three distinct signals:

1. **Image Context:** The *global image context*, conditioned via Feature-wise Linear Modulation (FiLM) [42] on the highest-level ($i = 2$) correlations.
2. **Recurrent State:** The model's *last predictions* (positions, covariances, and update deltas), projected to 32 dimensions each and processed by a 2-layer MLP.
3. **Landmark Identity:** The spectral *query encodings*, processed by a separate 2-layer MLP (denoted as $\mathbf{q}_n$).

These three signals are concatenated and compressed via a 2-layer MLP before RMS normalization, giving the resulting context vector $\mathbf{c}_n$. We residual-add a projected version of $\mathbf{c}_n$ to the correlation features to obtain a unified feature vector $\mathbf{f}_n$ for each landmark n. Both $\mathbf{f}_n$ and $\mathbf{c}_n$ are 256-dimensional.

3.1.5. Recurrent Optimization

The recurrent optimization module closes the iterative refinement loop. It maintains a per-landmark hidden state $\mathbf{h}_n \in \mathbb{R}^{128}$ that persists both across refinement iterations within a frame and across consecutive frames, carrying temporal semantics forward in time. Each update step first exchanges information across landmarks in a Write-Mix-Read block (Figure 2) and then integrates the result into the hidden state through a GRU update.

Write-Mix-Read. WMR factorizes cross-landmark communication through $S = 8$ *basis slots* per head (4 heads of width $D = 64$), reducing the interaction to a low-rank

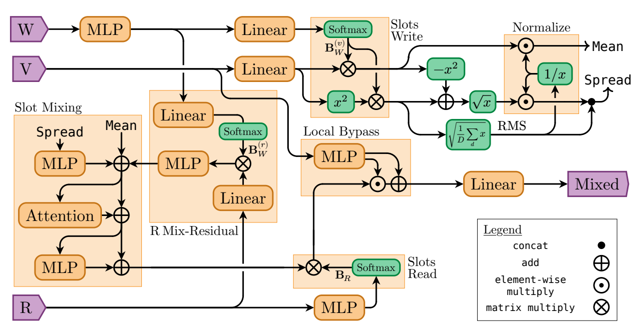

Figure 2. Write-Mix-Read block for spatio-temporal fusion.

mapping with $O(N)$ runtime versus $O(N^2)$ for full attention. Routing is *asymmetric*. The stable context features $W = \mathbf{c}_n$ decide what each landmark *writes*, while the hidden-state/landmark-identity pairs $R = [\mathbf{h}'_n, \mathbf{q}_n]$, with $\mathbf{h}'_n = \mathrm{RMSNorm}(\mathbf{h}_n)$, decide what each landmark *reads*; the correlation features $V = \mathbf{f}_n$ are the *values* that are written, mixed, and read back.

Concretely, the context features produce two write bases $\mathbf{B}_W^{(v)}, \mathbf{B}_W^{(r)} \in \mathbb{R}^{S \times N}$ with elements b_{sn}, learned-temperature-scaled and softmax-normalized *over landmarks*. Given projected correlation and read features $\mathbf{v}_n, \mathbf{r}_n \in \mathbb{R}^D$, each slot accumulates weighted first and second moments,

$$\begin{aligned} \boldsymbol{\mu}_s &= \sum_n b_{sn}^{(v)} \mathbf{v}_n,\ \mathbf{m}_s^{(2)} = \sum_n b_{sn}^{(v)} \mathbf{v}_n^2,\ \tilde{\mathbf{r}}_s = \sum_n b_{sn}^{(r)} \mathbf{r}_n, \\ \boldsymbol{\sigma}_s &= \sqrt{\mathbf{m}_s^{(2)} - \boldsymbol{\mu}_s^2}, \quad \mathrm{RMS}_s = \sqrt{\frac{1}{D} \mathbf{1} \cdot \mathbf{m}_s^{(2)}}. \end{aligned} \tag{4}$$

Conceptually, a basis slot can be viewed as a high-dimensional soft cluster of landmark features. The write basis softly assigns landmarks to S overlapping groups with centroid $\boldsymbol{\mu}_s$ and dispersion $\boldsymbol{\sigma}_s$. The dispersion lets a slot encode the agreement of its members. Normalizing both quantities by the RMS decomposes the slot energy into coherent and dispersed components, $\hat{\boldsymbol{\mu}}_s = \boldsymbol{\mu}_s / \mathrm{RMS}_s$ and $\hat{\boldsymbol{\sigma}}_s = \boldsymbol{\sigma}_s / \mathrm{RMS}_s$, so that $\sum_d (\hat{\mu}_{sd}^2 + \hat{\sigma}_{sd}^2) = D$.

Before readback, each slot is refined by four residual branches gated by a learned per-head scalar initialized near zero [43, 44]. (i) A Gated Linear Unit (GLU) [45] projection of the dispersion descriptor is added to $\boldsymbol{\mu}_s$, allowing the slot to be reweighted by the agreement of its members. (ii) The GLU-projected read centroid $\tilde{\mathbf{r}}_s$ is added to the slot. This is the pathway through which temporal semantics enter. Each slot is conditioned on the recurrent states of the landmarks routed to it, so the temporal history of one cluster can modulate the content read back by another. (iii) Self-attention across S slots, followed by (iv) a 2-layer MLP lets the learned topology factors exchange information. We denote the slot results by $\tilde{\mathbf{v}}_s \in \mathbb{R}^{1 \times D}$ stacked as $\tilde{\mathbf{V}} \in \mathbb{R}^{S \times D}$.

The read side mirrors the write side. State/identity-pairs yield a read basis $\mathbf{B}_R \in \mathbb{R}^{N \times S}$, softmax-normalized *over slots*. Each landmark reads a convex combination $\mathbf{Z}^{(r)} = \mathbf{B}_R \tilde{\mathbf{V}} \in \mathbb{R}^{N \times D}$ of the processed slots. Composing write and read (before slot refinement) gives a directed interaction matrix $\mathbf{A} = \mathbf{B}_R \mathbf{B}_W^{(v)}$ of rank at most S, a content-based, low-rank approximation of full landmark attention. Since all per-landmark quantities are bound to corresponding query points and all maps are shared, WMR is *equivariant* to landmark permutations, and the slots (within a given query set) *invariant*.

The output integrates a local bypass via FiLM of the correlation features, $\mathbf{z}_n^{(l)} = \mathbf{z}_n^{(r)} \odot \lambda(\mathbf{f}_n) + \beta(\mathbf{f}_n)$ with λ initialized to one and β to near zero, so the bypass starts off inactive. Finally, all parallel heads h are concatenated and out-projected, $\mathbf{z}_n = \mathbf{P}_o[\mathbf{z}_{nh}^{(l)}, ...]$. During training, slot-writing is randomly disabled for 15% of landmarks (entirely for pupil landmarks, whose motion is largely independent of the rest of the face), with gradients back into the basis slots also stopped. Affected landmarks still read normally.

Gated State Update. The WMR output is split into candidate features and gate features. The latter produce three sigmoid gates—reset $\mathbf{r}^{(g)}$, update $\mathbf{u}^{(g)}$, and gain $\mathbf{g}^{(g)}$—which form the candidate and the next hidden state,

$$\begin{aligned} \tilde{\mathbf{h}}_n &= \tanh\!\Big(\mathbf{r}_n^{(g)} \odot \mathbf{P}_h \mathbf{h}'_{n,t-1} + \mathbf{z}_n\Big) \odot \mathbf{g}_n^{(g)}, \\ \mathbf{h}_{nt} &= \Big(\mathbf{1} - \mathbf{u}_n^{(g)}\Big) \odot \mathbf{h}_{n,t-1} + \mathbf{u}_n^{(g)} \odot \tilde{\mathbf{h}}_n. \end{aligned} \tag{5}$$

t indexes successive refinement steps, including those that cross frame boundaries. This setup mirrors a GRU, except that the input map is absorbed into the WMR output projection. The added gain gate scales the bounded candidate multiplicatively, independent of the tanh argument, so unreliable candidates can be suppressed without saturating the nonlinearity. Gates are initialized toward state preservation. The update gate starts at $\mathrm{logit}(0.4)$, the gain gate at $\mathrm{logit}(0.9)$, while the reset gate begins centered at $\mathrm{logit}(0.5)$. The new hidden state is mapped by a two-headed GLU projection into separate coordinate and covariance streams; linear heads then emit the position delta $[\Delta x, \Delta y]$ and three covariance parameters. $[x, y]_t = [x_{t-1} + \Delta x, y_{t-1} + \Delta y]$. We use normalized image coordinates with $x, y \in [-1, 1]$.

Covariance Parameterization. Extending the scalar variance of [16, 17] to a full 2D covariance $\boldsymbol{\Sigma}$, the head predicts $\boldsymbol{\theta} = \big[\log \sigma_x, \log \sigma_y, \mathrm{artanh}(\rho)\big]$, such that

$$\boldsymbol{\Sigma} = \begin{pmatrix} \sigma_x^2 & \rho \sigma_x \sigma_y \\ \rho \sigma_x \sigma_y & \sigma_y^2 \end{pmatrix},\ \mathbf{L} = \begin{pmatrix} \sigma_x & 0 \\ \rho \sigma_y & \sigma_y \sqrt{1 - \rho^2} \end{pmatrix}. \tag{6}$$

Compared to $[\log l_{11}, \log l_{22}, l_{21}]$, a Cholesky parameterization with $\boldsymbol{\Sigma} = \mathbf{L}\mathbf{L}^\intercal$—where $\sigma_y^2 = l_{21}^2 + l_{22}^2$ couples the marginal variance with the correlation and log/linear domains are mixed—in the $\sigma\rho$-parameterization, all parameters are in the log-domain and decoupled. $2\,\mathrm{artanh}(\rho) = \mathrm{logit}((1 + \rho)/2)$ is the log-odds of the correlation rescaled to $(0, 1)$. Positive definiteness follows from $\sigma_x, \sigma_y > 0$ and $|\rho| < 1$.

3.2. Training Procedure

QLOT is trained by unrolling the model over video clips of length L. Frame $l \in \{1, ..., L\}$ is refined over $i \in \{1, ..., I\}$ iterations, yielding a total of $T = L \cdot I$ refinement steps indexed by $t(l, i) = (l - 1) \cdot I + i$. Between frames, the coordinate estimate $\boldsymbol{\mu}$, covariance parameters $\boldsymbol{\theta}$, and hidden state $\mathbf{h}$ from the final iteration ($i = I$) of frame l initialize the first iteration of frame $l + 1$. Labels are denoted $\mathbf{y}$.

To train for both cold-start detection and single-iteration steady-state tracking, the per-frame iteration budget I is sampled dynamically: With a 0.1% chance, set $I = 12$. Otherwise, steps < 2,000: $I \sim \mathcal{U}(2, 6)$; steps 2,000 to 4,999:

$I \sim \mathcal{U}(2,5)$; steps 5,000 to 9,999: $I \sim \mathcal{U}(2,4)$; steps $\geq$ 10,000: $I \sim \mathcal{U}(2,3)$ with 5% of clips trained at $I = 1$. We reset $\boldsymbol{\mu}$, $\boldsymbol{\theta}$ and $\mathbf{h}$ with probability 0.5% to train failure recovery. After all iterations of each frame we clamp $-10 \leq \log(\sigma_x)$, $\log(\sigma_y) \leq 3$ and $-2 \leq x, y \leq 2$ for numerical stability. Furthermore, we stop gradients to the previous frame's coordinates $\boldsymbol{\mu}_{t-1}$ as suggested by [15], but keep $\boldsymbol{\theta}_{t-1}$ live. Initial coordinates are set to randomly perturbed $[x, y]$ query point positions, $\boldsymbol{\theta}$ and $\mathbf{h}$ are initialized to zero.

We limit training to 110 epochs (1,000 steps per epoch) using AdamW (weight decay 0.005, gradient norm clipping at 1.0) with a cosine annealing learning rate schedule from $\eta_0 = 2 \cdot 10^{-4}$ to $\eta_{\min} = 10^{-5}$. The HGNetV2 backbone is frozen for the first 400 steps.

3.2.1. Datasets and Video Synthesis

Each step we sample from all four datasets:

- **FaceSynthetics** [46]: 95,000 training, 2,500 validation, and 2,500 test synthetic images with 70 landmarks.
- **WFLW** [47]: 7,000 training and 500 validation images from the training split, evaluated on the standard test sets with 98 landmarks.
- **WFLW-V** [5]: 1,000 video sequences (800 training, 50 validation, 150 test) with 98 landmarks. Training videos are sampled with a random stride $\sim \mathcal{U}(1, 10)$ as $L = 16$ clips.
- **300-W** [48]: 2,848 training and 300 validation images with 68 landmarks, evaluated on the standard common and challenging test sets.

Face Crop Generation. Input crops are centered on the landmark labels' bounding box and padded by a dataset-specific ratio (75% for FaceSynthetics, 35% for WFLW, 40% for WFLW-V, 55% for 300-W), then expanded to a square aspect ratio and augmented by a random affine per clip. Validation and test sets use a fixed padding ratio of 10%. Images are resized to 224×224 pixels after augmentations.

Clip Generation. Static images are converted into synthetic clips of length $L = 4$. Correlated affine transformations (translation, scale, rotation, shear) synthesize video motion. Photometric augmentations (color jitter, brightness/contrast, gamma, CLAHE, plasma shadows, sun flares) are sampled once per clip and applied to all frames, while noise (Gaussian, ISO, shot) and blur (Gaussian, motion) are applied per frame.

3.2.2. Loss Function

We use a composite loss of five terms, $\mathcal{L} = \mathcal{L}_{\text{pos}} + \mathcal{L}_{\text{cov}} + 0.5\mathcal{L}_{\text{delta}} + 0.25\mathcal{L}_{\text{acc}} + \lambda(s)\mathcal{L}_{\text{hcons}}$. Where defined, each loss is averaged over landmarks n, iterations i, and frames l.

Spatial accuracy is supervised by an L_2 loss with exponential decay weights $w_i = 0.85^{I-i}$ and $W = \sum_i w_i$,

$$\mathcal{L}_{\text{pos}} = \frac{w_i}{W} \|\boldsymbol{\mu}_{lin} - \mathbf{y}_{ln}\|_2. \tag{7}$$

Spatial uncertainty is trained via Gaussian Negative Log-Likelihood (GNLL),

$$\text{GNLL}(\boldsymbol{\mu}, \boldsymbol{\Sigma}, \mathbf{y}) = \frac{1}{2} \log \det(\boldsymbol{\Sigma}) + \frac{1}{2} \mathbf{x}^{\mathsf{T}} \boldsymbol{\Sigma}^{-1} \mathbf{x}, \tag{8}$$

where $\mathbf{x} = \boldsymbol{\mu} - \mathbf{y}$. The coordinate residual is detached (sg[·]),

$$\mathcal{L}_{\text{cov}} = \text{GNLL}(\text{sg}[\boldsymbol{\mu}_{lin}], \boldsymbol{\Sigma}_{lin}, \mathbf{y}_{ln}). \tag{9}$$

We split coordinate and uncertainty supervision into disjoint terms. Jointly trained heteroscedastic NLL couples the mean and variance gradients through $\boldsymbol{\Sigma}^{-1}$, which permits poor but stable equilibria where the covariance absorbs the residual instead of the estimate being corrected [49]. Stopping the residual's gradients makes $\mathcal{L}_{\text{cov}}$ a pure calibration objective for $\boldsymbol{\Sigma}$, while $\mathcal{L}_{\text{pos}}$ trains accuracy mostly unaffected by $\boldsymbol{\Sigma}^{-1}$.

Temporal GNLL terms supervise the trajectory against the final iteration of previous frames. Covariances are detached and used only as inverse-variance weights, allowing us to assume uncorrelated errors between frames. The *delta* term matches first differences $\Delta\boldsymbol{\mu}_{li} = \boldsymbol{\mu}_{li} - \boldsymbol{\mu}_{l-1,I}$ and $\Delta\mathbf{y}_l = \mathbf{y}_l - \mathbf{y}_{l-1}$ to $\boldsymbol{\Sigma}^{(v)}_{li} = \boldsymbol{\Sigma}_{li} + \boldsymbol{\Sigma}_{l-1,I}$,

$$\mathcal{L}_{\text{delta}} = \text{GNLL}\Big(\Delta\boldsymbol{\mu}_{lin}, \text{sg}\Big[\boldsymbol{\Sigma}^{(v)}_{lin}\Big], \Delta\mathbf{y}_{ln}\Big). \tag{10}$$

The *acceleration* term matches $\Delta^2\boldsymbol{\mu}_{li} = \Delta\boldsymbol{\mu}_{li} - \Delta\boldsymbol{\mu}_{l-1,I}$ (and likewise $\mathbf{y}$) to $\boldsymbol{\Sigma}^{(a)}_{li} = \boldsymbol{\Sigma}_{li} + 4\boldsymbol{\Sigma}_{l-1,I} + \boldsymbol{\Sigma}_{l-2,I}$,

$$\mathcal{L}_{\text{acc}} = \text{GNLL}\Big(\Delta^2\boldsymbol{\mu}_{lin}, \text{sg}\Big[\boldsymbol{\Sigma}^{(a)}_{lin}\Big], \Delta^2\mathbf{y}_{ln}\Big). \tag{11}$$

Horizontal consistency is enforced for WFLW and 300-W, which exhibit left-right annotation asymmetries. Batches contain interleaved original and horizontally flipped image pairs. Predictions on flipped images are mapped back to the original coordinate frame. The loss computes

$$\mathcal{L}_{\text{hcons}} = \left\|\Delta\boldsymbol{\mu}^{\text{flip}}_{lin}\right\|_2^2 + 0.1 \cdot \|\Delta\boldsymbol{\theta}_{lin}\|_1 \tag{12}$$

where $\Delta\boldsymbol{\mu}^{\text{flip}} = \boldsymbol{\mu} - \boldsymbol{\mu}'_{\text{flip}}$ (similarly for $\Delta\boldsymbol{\theta}$). The weight $\lambda(s)$ ramps linearly from 0 to 4.0 between training steps $s =$ 2,000 and 12,000.

3.3. Temporal Stability Metrics

While the Normalized Mean Error (NME) quantifies spatial accuracy, it is blind to the temporal behavior of the error. A jitter metric, the Normalized Mean Flicker (NMF) [5], is the frame-to-frame increment $\left\|\mathbf{e}_{ln} - \mathbf{e}_{l-1,n}\right\|_2$ of the landmark error $\mathbf{e}_{ln} = \boldsymbol{\mu}_{ln} - \mathbf{y}_{ln}$, normalized by the face size $d_l = \sqrt{h_l w_l}$ and aggregated in the RMS over landmarks and frames (mean over clips). NMF is a *precision* measure where all temporal change is penalized equally and constant offsets are ignored. As a first-difference statistic it cannot separate worst-case jitter, a sign-alternating error, from smooth drift.

We therefore generalize NMF to a multi-timescale stability profile based on the Allan variance [50]. With the normalized error $\tilde{\mathbf{e}}_{ln} = \mathbf{e}_{ln}/d_l$ and its block averages $\bar{\mathbf{e}}_{ln}(m) = \frac{1}{m}\sum_{i=l}^{l+m-1} \tilde{\mathbf{e}}_{in}$ over m consecutive frames, the

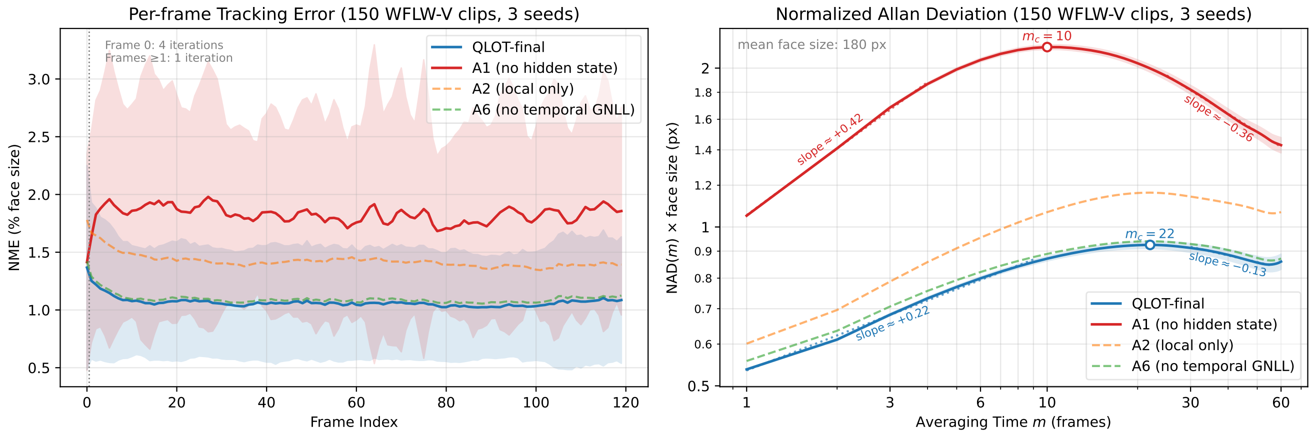

Figure 3. Left, per-frame WFLW-V NME with (QLOT-final) and without (A1) recurrent state carry; right, the corresponding NAD curves; A2 and A6 shown for comparison. QLOT-final has much lower jitter and drift. Whereas A1 integrates freely (slope $+0.42$ = free random walk for $m \in [1, 10]$), QLOT-final's slower rise ($+0.22$) and later peak ($m_c = 22 > 10$) reveal memory-bounded, self-correcting dynamics.

normalized Allan variance compares successive blocks,

$$\sigma^2(m) = \frac{1}{2(L-2m+1)} \sum_{l=1}^{L-2m+1} \left\| \bar{\mathbf{e}}_{l+m}(m) - \bar{\mathbf{e}}_l(m) \right\|_2^2. \tag{13}$$

Then, the *Normalized Allan Deviation* (NAD) is the RMS of $\sigma(m)$ over landmarks (mean over clips). $\text{NAD}(1) \approx \text{NMF}/\sqrt{2}$, up to normalization convention. NMF is thus the shortest-timescale point of the NAD curve, and its conflation of jitter and drift becomes precise: For the simple model $\tilde{\mathbf{e}}_l = \mathbf{j}_l + \mathbf{d}_l$ where $\text{NAD}^2(m) = \sigma_j^2/m + m^2\|\mathbf{v}\|^2/2$ with zero-mean white jitter $\mathbf{j}_l$ of per-frame variance σ_j^2 and linear drift $\mathbf{d}_l = l \cdot \mathbf{v}$, block averaging attenuates the jitter contribution as σ_j^2/m, whereas linear drift survives. So, the NMF alone cannot distinguish between jitter and drift.

However, plotted against m on log-log axes, the NAD curve diagnoses the random processes composing the error (non-unique relation to power spectral density) [50]. Asymptotically, *white jitter* decays with slope $-1/2$, *long-range correlations* ($1/f$ noise) flatten the curve into the zero-slope *bias-instability* floor, an *error random walk* rises with slope $+1/2$, and *linear drift* with $+1$.

The type of detector needs to be distinguished. An absolute-coordinate predictor re-anchors to the image at every frame and has bounded drift by construction, whereas a delta predictor integrates delta errors $\tilde{\mathbf{u}}_l = \tilde{\mathbf{e}}_l - \tilde{\mathbf{e}}_{l-1}$ (exactly as measured by NMF) across frames. Uncorrelated delta noise then accumulates into an *error random walk*. A persistent delta bias drifts faster (slope $+1$), while anticorrelated delta noise—visual feedback correcting accumulated errors—flattens and ultimately bounds the drift. This produces a hump in the graph with the maximum at the error correlation time m_c. Then, a falling slope separates a zero-reverting ($-1/2$) error from the bias-instability floor (0 slope). See the supplement for further details.

4. Experiments

We evaluate QLOT-full (Section 3.) on static accuracy (WFLW, 300-W, FaceSynthetics) and video stability (WFLW-V) against pre-specified claims. Seven ablation variants that change one component of the model test each claim: On the **recurrent memory** (*A1*, hidden-state reset), the **update operator** (*A2*, parameter-matched local-only GRU; *A3*, standard GRU; *A4*, mean-only WMR), the **supervision recipe** (*A5*, L_2 only; *A6*, no temporal GNLL), and the **query encoding** (*A7*, geometric Fourier). All configurations are trained from scratch under an identical protocol, mean over 3 seeds. Checkpoints are selected on the validation sets by lowest static NME, then choosing lowest NMF among the best three. Detailed protocols, full ablation tables, and per-claim statistical analysis are deferred to the supplementary material. We report only the prominent results here.

4.1. Results

Recurrent memory (A1) is the dominant factor. Resetting the hidden state at every frame, while still carrying the previous coordinates and covariance, nearly doubles flicker (NMF 211.4 vs. 109.5, +93%), raises video NME by 68%, and elevates NAD at all timescales, at essentially unchanged static accuracy. Figure 3 shows why: The variants are indistinguishable at frame 0 (4 iterations each). The gap opens at frame 1—the first single-iteration frame—and persists for the rest of the clip. Thus, the hidden state is not a smoother applied to otherwise converged estimates, but what lets the tracker stay locked in one iteration per frame, and recover quickly when lost. Coordinate forwarding tells the model *where* to focus in the next frame, the hidden state carries *past information of the track*—without it, every frame starts cold.

Update operator (A2–A4). Removing cross-landmark communication (A2) is the most damaging architectural

Method, Year	Params	MACs	300-W (NME)		WFLW (NME)							Synth	WFLW-V			
			Com	Chal	Full	Pose	Expr	Ill	Mkp	Occl	Blur	NME	NME_E	NME_H	NMF_E	NMF_H
RwR [5], 2023	21.8 M	–	–	–	**3.92**	6.86	**3.94**	4.17	**3.75**	4.77	4.59	–	1.24	2.30	82.74	172.95
Inf3DLmks [7], 2024	>100 M*	–	2.89	5.71	–	–	–	–	–	–	–	–	–	–	–	–
TUFA [26], 2025	36.0 M	17.6 G	**2.59**	**4.45**	3.93	**6.48**	4.11	**3.82**	3.81	**4.68**	**4.53**	–	–	–	–	–
QLOT-final (ours)	**1.98 M**	**0.65 G**	3.03	5.37	3.93	6.65	4.07	3.88	3.90	4.75	4.60	1.62	**0.83**	**1.30**	**77.7**	**138.5**

Table 1. Comparison with published face-landmark methods. Static NME (%, inter-ocular) on 300-W, WFLW, and FaceSynthetics (%, face-size). WFLW-V video NME (%, face-size) and NMF on our test split (easy, hard), each video truncated to 120 frames. MACs (Multiply-Accumulate operations) are the per-frame tracking cost. [–] = not reported. *Estimated from paper.

Figure 4. Qualitative results with predicted 95% confidence ellipses.

change after A1 (NMF +10.5%, video NME +31%, WFLW NME +8.7%, significant in every seed). Landmarks do not move independently, and the mixer exploits this structure. Our GRU formulation is at least as good as a standard GRU variant (A3). Removing the WMR dispersion descriptor (A4) degrades nothing and slightly *improves* stability (NMF −1.3%, video NME −2.1%, both significant). The routed temporal states may carry similar information, and variance estimation is itself noise-prone. We adopt the simpler mean-only WMR (1.7% fewer parameters) and call it *QLOT-final*.

Supervision (A5, A6) and query encoding (A7). Dropping the temporal GNLL terms (A6) raises NMF by 2.4% and NAD mostly at short timescales (Figure 3), with static NME unchanged. A pure L_2 loss (A5) is nearly identical (+2.6%) to this. So, the temporal terms account for essentially all of the supervision gain, an effect that is real but an order of magnitude smaller than state carry. PMPE (A7) is a null result on all metrics and is retained for flexibility. Checkpoint selection is itself a methodological variable. Purely NME-based selection yields worse trackers (NMF +0.8%, video NME +1.9%) at identical static accuracy, so we recommend reporting the selection criterion explicitly in video-landmark work.

Comparison to prior work. Table 1 compares QLOT-final with three published methods: At 1.98M parameters and 0.65G MACs per tracked frame (1 iteration, 98 landmarks), QLOT uses an order of magnitude fewer parameters than prior work at competitive static accuracy. It runs at up to 680 FPS on a single NVIDIA RTX 3080 GPU, where it shows almost linear inference cost in the number of landmarks tracked. Figure 4 displays some annotated images.

Against RwR [5], the strongest published video baseline, itself already below every EMA-smoothed detector it was compared against, QLOT lowers NMF by 15% overall and by 20% on the hard split. The test protocol is near-identical. While RwR calculates jitter on the whole WFLW-V dataset, as they can only train on images, we evaluate on our test subset (75 hard, 75 easy videos). See the supplement for a full protocol discussion.

5. Conclusion

We presented QLOT, a recurrent architecture that formulates face landmark tracking as query-conditioned learned optimization. A per-landmark hidden state carried across frames and exchanged, in addition to RAFT-like correlation features, through a Write-Mix-Read block keeps the tracker locked with a single iteration per frame. Ablations identify recurrent state carry as the dominant source of temporal stability, with temporal GNLL supervision adding a smaller gain. Canonical 3D query points enable joint training on heterogeneous image and video data. At 1.98M parameters and 0.65G MACs per frame, QLOT-final achieves competitive-to-SOTA static accuracy while reducing flicker by 15% over the strongest video baseline, running at up to 680 FPS on a consumer GPU. The proposed Normalized Allan Deviation generalizes NMF to a multi-timescale profile that separates jitter from drift, providing a principled stability diagnostic for future tracking research.

References

[1] Ivan Grishchenko, Artsiom Ablavatski, Yury Kartynnik, Karthik Raveendran, and Matthias Grundmann. Attention Mesh: High-fidelity face mesh prediction in real-time. In *CVPR Workshop on Computer Vision for Augmented and Virtual Reality*, 2020. 1

[2] Shi Yin, Shijie Huang, Shangfei Wang, Jinshui Hu, Tao Guo, Bing Yin, Baocai Yin, and Cong Liu. 1DFormer: A transformer architecture learning 1D landmark representations for facial landmark tracking. In *IJCAI*, pages 1588–1597, 2024. 1, 2

[3] Seongmin Lee, Hyunse Yoon, Jiwoo Kang, Jungsu Kim, Jiwan Son, Jungwoo Huh, and Sanghoon Lee. Video-based stabilized 3D face alignment using temporal multi-discrimination. In *MMSP*, pages 1–6, 2023. 1

[4] Seongmin Lee, Hyunse Yoon, Sohyun Park, Sanghoon Lee, and Jiwoo Kang. Stabilized temporal 3D face alignment using landmark displacement learning. *Electronics*, 12(17):3735, 2023. 1

[5] Paul Micaelli, Arash Vahdat, Hongxu Yin, Jan Kautz, and Pavlo Molchanov. Recurrence without recurrence: Stable video landmark detection with deep equilibrium models. In *CVPR*, pages 22814–22825, 2023. 1, 2, 6, 8

[6] Hongyu Zhu, Hao Liu, Congcong Zhu, Zongyong Deng, and Xuehong Sun. Learning spatial-temporal deformable networks for unconstrained face alignment and tracking in videos. *Pattern Recognition*, 107:107354, 2020. 1, 2

[7] P. Chandran, G. Zoss, P. Gotardo, and D. Bradley. Infinite 3D landmarks: Improving continuous 2D facial landmark detection. *Computer Graphics Forum*, 43(6):e15126, 2024. 1, 2, 8

[8] Muhammad Arief Nugroho, Maman Abdurohman, Bayu Erfianto, Mahmud Dwi Sulistiyo, and Anita Ahmad Kasim. Comparative analysis of filtering techniques in eye landmark tracking: Kalman, Savitzky-Golay, and Gaussian. In *ICECCE*, pages 1–6, 2023. 1, 2

[9] Bing-Fei Wu, Bo-Rui Chen, and Chun-Fei Hsu. Design of a facial landmark detection system using a dynamic optical flow approach. *IEEE Access*, 9:68737–68745, 2021. 1, 2

[10] Utsav Prabhu, Keshav Seshadri, and Marios Savvides. Automatic facial landmark tracking in video sequences using Kalman filter assisted active shape models. In *Trends and Topics in Computer Vision*, pages 86–99, 2012. 1, 2

[11] Xuanyi Dong, Shoou-I Yu, Xinshuo Weng, Shih-En Wei, Yi Yang, and Yaser Sheikh. Supervision-by-Registration: An unsupervised approach to improve the precision of facial landmark detectors. In *CVPR*, pages 360–368, 2018. 1

[12] Kai Su and Xin Geng. Soft facial landmark detection by label distribution learning. In *AAAI*, pages 5008–5015, 2019. 1

[13] Seongmin Lee, Hyunse Yoon, Sanghoon Lee, and Jiwoo Kang. Temporal facial alignment with triple discriminators. *Journal of Multimedia Information System*, 10:101–108, 2023. 1

[14] Xin Jin and Xiaoyang Tan. Face alignment in-the-wild: A survey. *Computer Vision and Image Understanding*, 162:1–22, 2017. 1, 2

[15] Zachary Teed and Jia Deng. RAFT: Recurrent All-pairs field transforms for optical flow. In *ECCV*, pages 402–419, 2020. 2, 4, 6

[16] Prashanth Chandran, Gaspard Zoss, Paulo Gotardo, and Derek Bradley. Continuous landmark detection with 3D queries. In *CVPR*, pages 16858–16867, 2023. 2, 3, 5

[17] Erroll Wood, Tadas Baltrušaitis, Charlie Hewitt, Matthew Johnson, Jingjing Shen, Nikola Milosavljević, et al. 3D face reconstruction with dense landmarks. In *ECCV*, pages 160–177, 2022. 2, 5

[18] Jagmohan Meher, Hector Allende-Cid, and Torbjörn E. M. Nordling. A survey and classification of face alignment methods based on face models. arXiv preprint arXiv:2311.03082, 2023. 2

[19] Arnaud Dapogny, Matthieu Cord, and Kevin Bailly. DeCaFA: Deep convolutional cascade for face alignment in the wild. In *ICCV*, pages 6892–6900, 2019. 2

[20] Chong-Yang Xiang, Jun-Yan He, Zhi-Qi Cheng, Xiao Wu, and Xian-Sheng Hua. POPoS: Improving efficient and robust facial landmark detection with parallel optimal position search. In *AAAI*, 2025. 2

[21] Xing Lan, Qinghao Hu, and Jian Cheng. HIH: Towards more accurate face alignment via heatmap in heatmap. In *CoRR*, 2021. 2

[22] Xinyao Wang, Liefeng Bo, and Li Fuxin. Adaptive wing loss for robust face alignment via heatmap regression. In *ICCV*, 2019. 2

[23] Ximin Sun, Yong Li, Pingjuan Niu, Jiangkai Jia, Yi Hao, Lu Chen, Shuzhen Han, Mingda Wang, and Yanjun Cui. 3D facial landmark detection techniques: A survey. In *EEET*, pages 53–58, 2024. 2

[24] George Trigeorgis, Patrick Snape, Mihalis A. Nicolaou, Epameinondas Antonakos, and Stefanos Zafeiriou. Mnemonic descent method: A recurrent process applied for End-to-end face alignment. In *CVPR*, pages 4177–4187, 2016. 2

[25] Kyunghyun Cho, Bart van Merriënboer, Caglar Gulcehre, Dzmitry Bahdanau, Fethi Bougares, Holger Schwenk, and Yoshua Bengio. Learning phrase representations using RNN Encoder–decoder for statistical machine translation. In *EMNLP*, pages 1724–1734, 2014. 2

[26] Jiahao Xia, Min Xu, Wenjian Huang, Jianguo Zhang, Haimin Zhang, and Chunxia Xiao. Mitigating knowledge discrepancies among multiple datasets for task-agnostic unified face alignment. *International Journal of Computer Vision*, 133(11):7985–8005, 2025. 2, 4, 8

[27] Jiahao Xia, Weiwei Qu, Wen-Fong Huang, Jianguo Zhang, Xi Wang, and Min Xu. Sparse local patch transformer for robust face alignment and landmarks inherent relation learning. In *CVPR*, pages 4042–4051, 2022. 2

[28] Jinpeng Li, Haibo Jin, Shengcai Liao, Ling Shao, and Pheng-Ann Heng. RePFormer: Refinement pyramid transformer for robust facial landmark detection. In *IJCAI*, pages 1088–1094, 2022. 2

[29] Ying Tai, Yicong Liang, Xiaoming Liu, Lei Duan, Jilin Li, Chengjie Wang, Feiyue Huang, and Yu Chen. Towards highly accurate and stable face alignment for High-resolution videos. In *AAAI*, pages 8893–8900, 2019. 2

[30] Jianzhu Guo, Xiangyu Zhu, Yang Yang, Fan Yang, Zhen Lei, and Stan Z Li. Towards fast, accurate and stable 3D dense face alignment. In *ECCV*, 2020. 2

[31] Libing Zeng, Lele Chen, Wentao Bao, Zhong Li, Yi Xu, Junsong Yuan, and Nima K. Kalantari. 3D-aware facial landmark detection via multi-view consistent training on synthetic data. In *CVPR*, pages 12747–12758, 2023. 2

[32] Shi Yin, Shangfei Wang, Xiaoping Chen, Enhong Chen, and Cong Liang. Attentive One-dimensional heatmap regression for facial landmark detection and tracking. In *ACM MM*, pages 538–546, 2020. 2

[33] Sinong Wang, Belinda Z. Li, Madian Khabsa, Han Fang, and Hao Ma. Linformer: Self-attention with linear complexity. arXiv preprint arXiv:2006.04768, 2020. 3

[34] Juho Lee, Yoonho Lee, Jungtaek Kim, Adam R. Kosiorek, Seungjin Choi, and Yee Whye Teh. Set transformer: A framework for attention-based Permutation-invariant neural networks. In *PMLR*, pages 3744–3753, 2019. 3

[35] Francesco Locatello, Dirk Weissenborn, Thomas Unterthiner, Aravindh Mahendran, Georg Heigold, Jakob Uszkoreit, Alexey Dosovitskiy, and Thomas Kipf. Object-centric learning with slot attention. In *NeurIPS*, pages 11525–11538, 2020. 3

[36] Andrew Jaegle, Sebastian Borgeaud, Jean-Baptiste Alayrac, Carl Doersch, Catalin Ionescu, David Ding, et al. Perceiver IO: A general architecture for structured inputs & outputs. In *ICLR*, 2022. 3

[37] PaddlePaddle Authors. PP-HGNetV2. https://github.com/PaddlePaddle/PaddleClas/blob/v2.6.0/docs/zh_CN/models/ImageNet1k/PP-HGNetV2.md, 2024. 3

[38] Tsung-Yi Lin, Piotr Dollár, Ross Girshick, Kaiming He, Bharath Hariharan, and Serge Belongie. Feature pyramid networks for object detection. In *CVPR*, pages 936–944, 2017. 3

[39] Ruilong Li, Karl Bladin, Yajie Zhao, Chinmay Chinara, Owen Ingraham, Pengda Xiang, et al. Learning formation of Physically-based face attributes. In *CVPR*, pages 3407–3416, 2020. 3

[40] Foundation AI Team, Kiran Bhat, Nishchaie Khanna, Karun Channa, Tinghui Zhou, Yiheng Zhu, et al. Cube: A Roblox view of 3D intelligence. arXiv preprint arXiv:2503.15475, 2025. 4

[41] Minghang Zhao, Shisheng Zhong, Xuyun Fu, Baoping Tang, and Michael Pecht. Deep residual shrinkage networks for fault diagnosis. *IEEE TII*, 16(7):4681–4690, 2020. 4

[42] Ethan Perez, Florian Strub, Harm de Vries, Vincent Dumoulin, and Aaron C. Courville. FiLM: Visual reasoning with a general conditioning layer. In *AAAI*, 2018. 4

[43] Thomas Bachlechner, Bodhisattwa Prasad Majumder, Henry Mao, Gary Cottrell, and Julian McAuley. ReZero is all you need: Fast convergence at large depth. In *Uncertainty in Artificial Intelligence*, pages 1352–1361, 2021. 5

[44] Hugo Touvron, Matthieu Cord, Alexandre Sablayrolles, Gabriel Synnaeve, and Hervé Jégou. Going deeper with image transformers. In *ICCV*, pages 32–42, 2021. 5

[45] Yann N. Dauphin, Angela Fan, Michael Auli, and David Grangier. Language modeling with gated convolutional networks. In *PMLR*, pages 933–941, 2017. 5

[46] Erroll Wood, Tadas Baltrušaitis, Charlie Hewitt, Sebastian Dziadzio, Thomas J. Cashman, and Jamie Shotton. Fake it till you make it: Face analysis in the wild using synthetic data alone. In *ICCV*, pages 3661–3671, 2021. 6

[47] Wayne Wu, Chen Qian, Shuo Yang, Quan Wang, Yici Cai, and Qiang Zhou. Look at boundary: A Boundary-aware face alignment algorithm. In *CVPR*, 2018. 6

[48] Christos Sagonas, Georgios Tzimiropoulos, Stefanos Zafeiriou, and Maja Pantic. 300 faces In-the-wild challenge: The first facial landmark localization challenge. In *ICCV Workshops*, pages 397–403, 2013. 6

[49] Maximilian Seitzer, Arash Tavakoli, Dimitrije Antic, and Georg Martius. On the pitfalls of heteroscedastic uncertainty estimation with probabilistic neural networks. In *ICLR*, 2022. 6

[50] Naser El-Sheimy, Haiying Hou, and Xiaoji Niu. Analysis and modeling of inertial sensors using Allan variance. *IEEE TIM*, 57:140–149, 2008. 6, 7